# Comprehensive Test Incident Report

Comprehensive Crash Analysis, 49 CFR Part 563 EDR Fusion, PyTorch AI Severity & Cryptographic Ledger

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| **Report ID:** REP-DS-20261024-3c84fa88-054713 | **Date/Time:** 2026-10-24 05:47:13 UTC |
| **Associated Case ID:** CASE_AF_8902 | **Lead Investigator:** Forensic Engineer #1024 |
| **Primary Evidence SHA-256:** `07f5e6619bc4937ebc98127ff89b...4f0324db` | **FIPS 180-4 Seal:** **VALIDATED & TAMPER-CHECKED** |

## 1. REGULATORY FRAMEWORK & EXECUTIVE SUMMARY

This digital forensic analysis report documents the data extraction, cryptographic verification, and kinematic reconstruction performed for **CASE_AF_8902**. Data extraction and audit logging comply with **Title 49 Code of Federal Regulations (CFR) Part 563 (Event Data Recorders)**, ISO/IEC 17025 laboratory verification principles, and FIPS 180-4 cryptographic standard. The authoritative forensic database is MySQL 8.0 (`autorecon`). Dataset ingestion verifies zero byte tampering relative to initial custody hash.

## 2. INCIDENT ENVIRONMENT & VEHICLE PARAMETERS

| | | | |
|---|---|---|---|
| **Subject Vehicle:** | 2024 Subject Vehicle Instrumented Test Unit | **VIN Number:** | `1G1FA1E38M0104821` |
| **Road Condition:** | Dry Asphalt | **Speed Limit:** | 65.0 km/h |
| **Initial Approach Speed:** | 65.0 km/h | **Calculated Delta-V:** | 28.5 km/h |

## 3. ARTIFICIAL INTELLIGENCE & MACHINE LEARNING INFERENCE

| Model Architecture | Prediction Output | Statistical Confidence | Training Baseline / Benchmark |
|---|---|---|---|
| PyTorch Deep Neural Net CrashSeverityNet (Dense+BatchNorm) | **SERIOUS INCAPACITATING** Injury Risk Score: 99.0 / 100 | 38.5% | NHTSA FARS 2024 National Census 8,157 Test Records (Acc: 59.00%) |
| Balanced Random Forest PatternClassifier (5-Class) | **FRONTAL COLLISION** | 94.4% | 54,374 Real Driver Crashes Balanced Class Weights (Acc: 76.51%) |

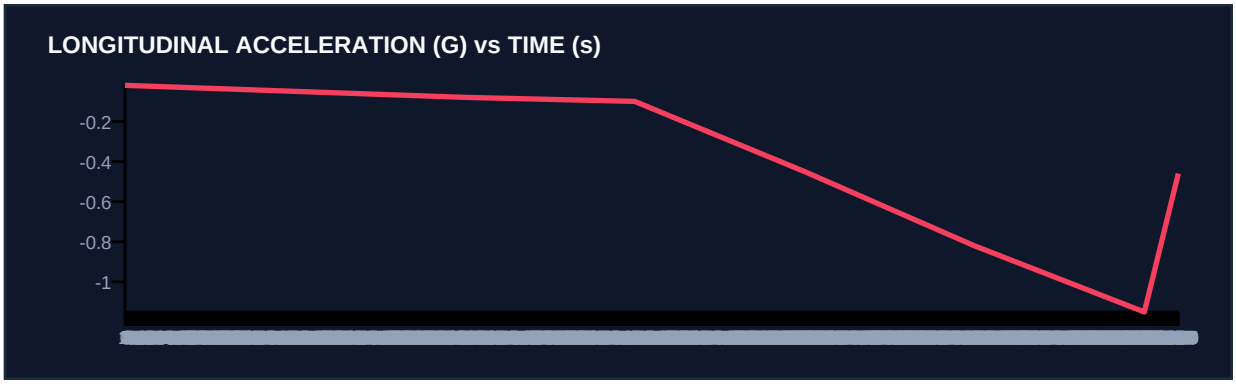
## 4. HISTORICAL PRECEDENT MATCHES (NHTSA FARS 2024 CENSUS)

| | | | | |
|---|---|---|---|---|
| 1 | ST_CASE #Census Record | **93.2%** | ANGLE | MINOR MODERATE INJURY |
| 2 | ST_CASE #Census Record | **93.2%** | ANGLE | SERIOUS INCAPACITATING |
| 3 | ST_CASE #Census Record | **93.2%** | ANGLE | FATAL CATASTROPHIC |

## 5. KINEMATIC TRAJECTORY & TELEMETRY CHARTS

Synchronized kinematic time series (-5.0s pre-impact to +2.5s post-impact):




## 6. FORENSIC TRUTH BOUNDARY & METHODOLOGY AUDIT

| Forensic Category | Channels / Parameters | Evidentiary Weight & Handling |
|---|---|---|
| **Recorded Facts** | File SHA-256, Ingestion Timestamp, File Byte Size, Header Fields | **High Evidentiary Value.** Bit-for-bit verifiable via cryptographic ledger. |
| **Derived Kinematics** | Delta-V (km/h), Deceleration Gs, Energy Loss | **Calculated.** Derived using Campbell crush energy and Newtonian impulse formulation. |
| **AI / ML Estimates** | KABCO Injury Class, Collision Pattern, Similarity % | **Probabilistic.** Neural network inference calibrated on NHTSA census data. |
| **Simulation Assumptions** | Visual Vehicle Geometry, 3D Mesh Orientation, Perspective Horizon | **Illustrative.** Physical approximation for investigator visualization. |

***MANDATORY FORENSIC DISCLAIMER:*** *This automated incident reconstruction report is generated by AUTO-RECON AI solely for investigative assistance, digital evidence preservation, and comparative crash research. Machine learning predictions and kinematic simulations represent probabilistic approximations and physical models; they do not constitute definitive proof of legal fault, proximate causation, or judicial culpability. Final engineering conclusions must be certified by an accredited professional forensic collision accident reconstructionist.*